

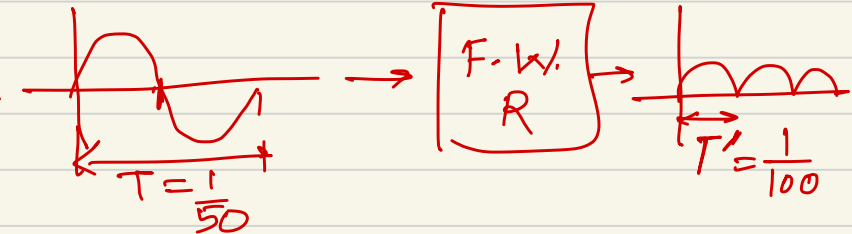
Mid-semester Exam

Stage 1: 1. In steady state, capacitor blocks DC, i.e. it acts as an open circuit.

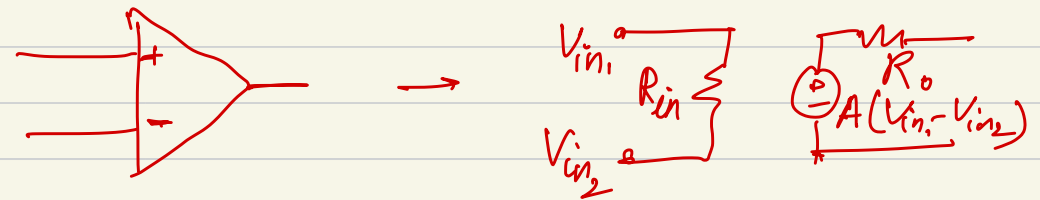
2. When the inductor is connected to a DC source, it will resist the change in the current.

Thus, if initial current is zero, the current will remain zero at the instant it is connected.

3. $\Rightarrow$ frequency of ripple voltage = 100 kHz.

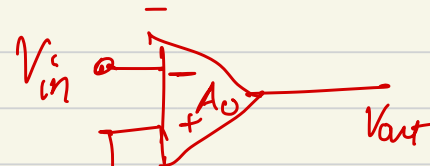


4. For ideal opamp,
 $R_{in} \rightarrow \infty$, $\Rightarrow$ current in input terminals is zero.



5. $V_{out} = -5V$

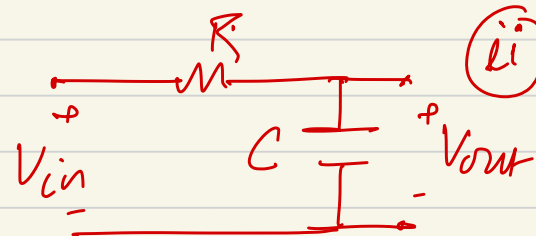
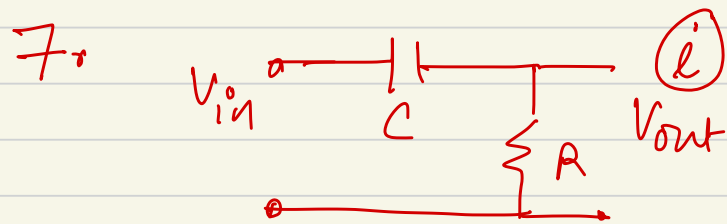
$$V_{in} = -\frac{V_{out}}{A_0} = \frac{5}{10^5} = 50 \mu V \approx 0 \text{ (virtual short)}$$



6. for inverting amplifier, $\frac{V_o}{V_i} = -\frac{R_f}{R_i} = -100$

$$\Rightarrow R_f = 100 R_i$$

if $V_i = 1V$, $V_o = -100V$, but V_o is going to be limited by DC supplies, $\Rightarrow \underline{V_o = -15V}$.



C behaves as an open circuit for low frequencies.

C " a short circuit for very high frequencies

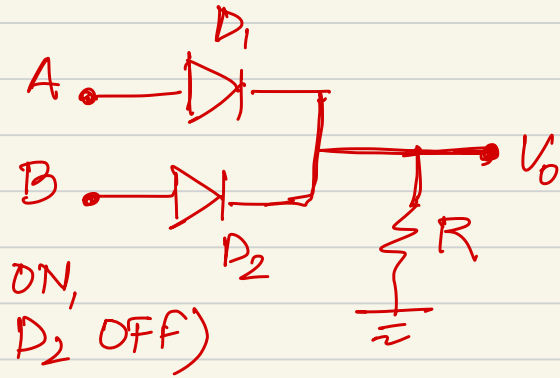
for (i), at low freq. $\rightarrow V_o = 0$, at high freq. $V_o \sim V_{in}$

ii) at low freq. $\rightarrow V_o \sim V_{in}$, at high freq. $V_o \sim 0$

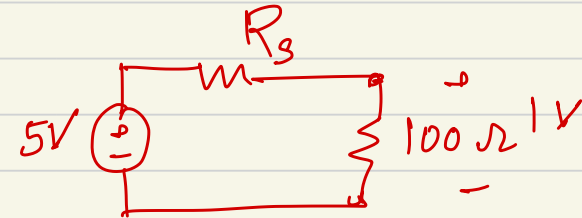
$\Rightarrow$ (ii) will give desired result.

8. OR gate, meaning if one of the input is '1' or 5V, output will be '1' i.e. 5V.

$A = 5V, B = 0V$ (D_1 ON, D_2 OFF)
 $\Rightarrow \underline{V_0 = 5V}$



9. the voltage drops because of finite input resistance of the sensor.

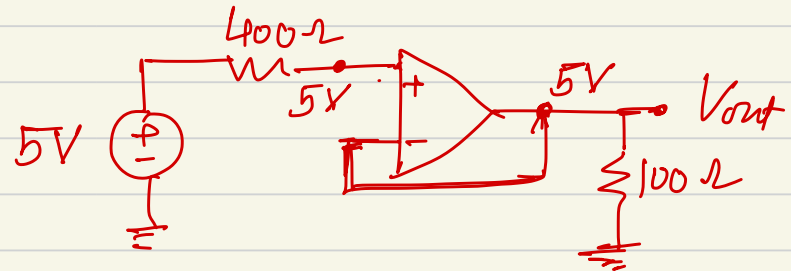


$$1 = \frac{100}{100 + R_s} \times 5$$

$$100 + R_s = 500$$

$$R_s = 400\Omega$$

10. As there is no current going into the ideal Opamp, the output voltage will be 5V.
 as $V_+ \approx V_- \Rightarrow V_{out} = 5V$

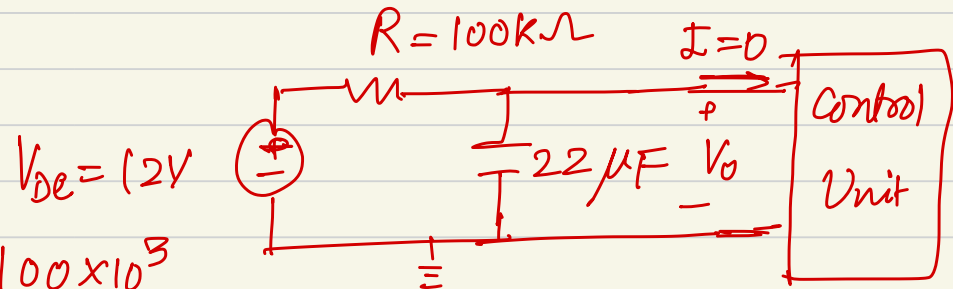


Stage 2 :-

Q11

A) time constant of the charging circuit

$$\tau = RC = 22 \times 10^{-6} \times 100 \times 10^3 \\ = 2.2 \text{ s}$$



B) If the current going inside control unit is zero, the time required to reach 9V.

$$V_0(t) = V_{De}(1 - e^{-t/\tau})$$

$$9 = 12(1 - e^{-t/\tau})$$

$$\Rightarrow \frac{3}{4} = 1 - e^{-t/\tau} \Rightarrow e^{-t/\tau} = \frac{1}{4}$$

$$\Rightarrow -\frac{t}{\tau} = \ln\left(\frac{1}{4}\right) \Rightarrow \boxed{t = \ln(4) \cdot \tau}$$

C) we need to recalculate the time constant of the circuit. For that we convert it using Thevenin.

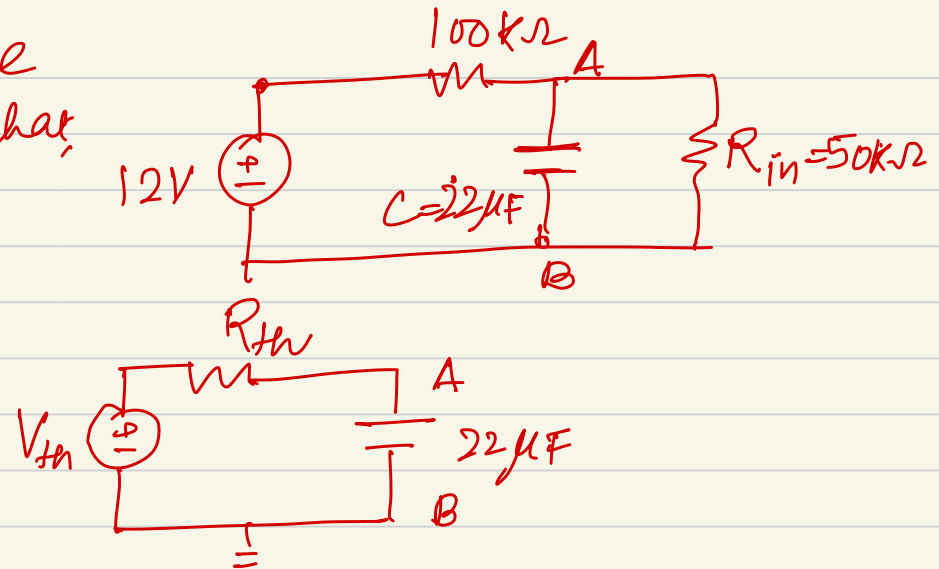
$$V_{th} = \frac{50k\Omega}{150k\Omega} \times 12 = 4V$$

$$R_{th} = \frac{50 \times 100k}{150k} = \frac{100k}{3} = 33.3k$$

$$\text{new } \tau = R_{th} C = 33.3k \times 22\mu F < 2.2s$$

it decreases.

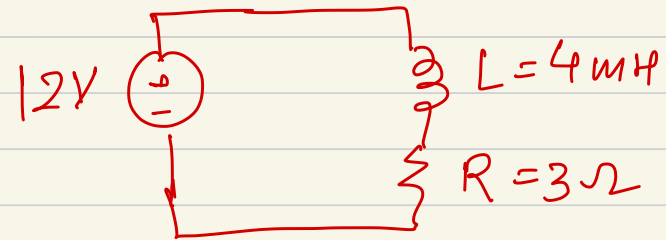
→ As $V_{th} = 4V$ (instead of $12V$ in case B), the robot will not turn on as the charged voltage can not exceed $4V$.
(need $9V$)



Q12.

- i) In steady state inductor will act as a short. Circuit.

$$i.e. \quad I = \frac{12}{3} = 4A$$

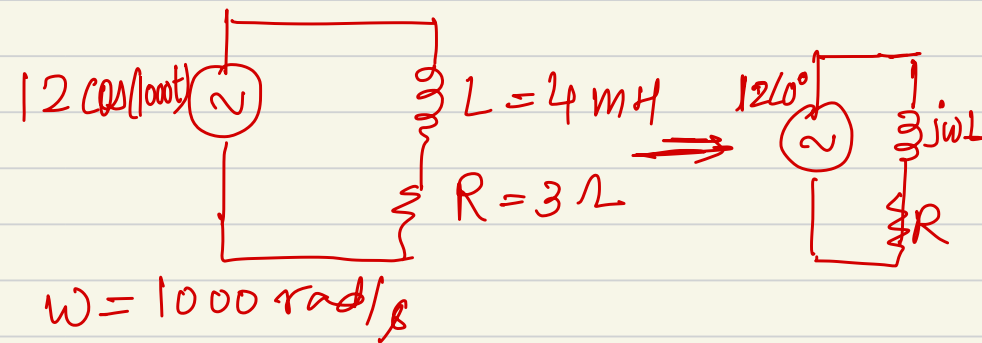


- If the supply is abruptly removed, the current goes to zero in $10\mu s$.

$$\begin{aligned} \Rightarrow \text{the voltage across the inductor} &= L \frac{di}{dt} \approx L \frac{\Delta i}{\Delta t} \\ &= 4 \times 10^{-3} \left[\frac{0 - 4A}{10\mu - 0} \right] \\ &= \frac{-16 \times 10^{-3}}{10 \times 10^{-6}} = -1.6 \text{ kV} \end{aligned}$$

- ii) To calculate AC current, we can represent the circuit in phasor domain.

$$I = \frac{12 \angle 0^\circ}{3 + j1000 \times 4 \times 10^{-3}} = \frac{12 \angle 0^\circ}{3 + j4}$$



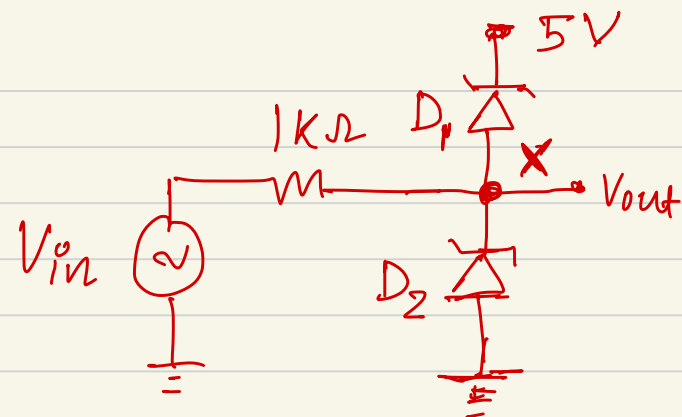
$$\Rightarrow I = \frac{12 \angle 0^\circ}{5 \angle \tan^{-1} 4/3} = 2.4 \angle -\tan^{-1} 4/3 \text{ A.}$$

$|I| = 2.4 \text{ A}$ is lesser than 4 A as the reactance due to L is finite, which results in higher impedance (i.e. 5Ω) compared to 3Ω in steady state.

Q13.

$$V_{in} = 20 \sin \omega t$$

$$V_D = 0.2V \text{ (in FB diode)}$$



i) When $V_{in} \rightarrow +ve$

D_1 will be ON when V_x crosses $5.2V$, D_1 will be OFF if $V_{in} \leq 5.2V$

D_2 will be OFF for the whole cycle. (open)

When $0 \leq V_{in} \leq 5.2V \Rightarrow D_1, D_2$ OFF

$$\Rightarrow V_{out} = V_{in}$$

$V_{out_{min}} = -0.2V$
$V_{out_{max}} = 5.2V$

When $5.2V < V_{in} \leq 20V \Rightarrow D_1$ ON, D_2 OFF
(Short)

$$V_{out} = 5.2V$$

When $V_{in} \rightarrow -ve$, D_1 will be OFF as $V_{D1} < -ve$

When V_{in} or $V_x \leq -0.2V$, D_2 will turn ON, i.e. $-0.2V \leq V_{in} < 0V$, D_2 will be OFF.

i.e. $-0.2V \leq V_{in} < 0$, D_1 , D_2 OFF

$$V_{out} = V_{in}$$

When $-20V \leq V_{in} < -0.2V$, D_1 OFF, D_2 ON

$$V_{out} = -0.2V$$

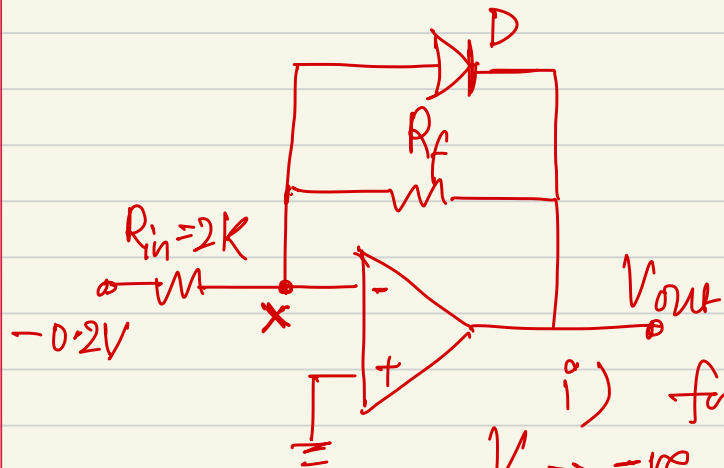
i) when $V_{in} \rightarrow +ve$, $V_{R_{max}} = V_{in_{max}} - V_{out} = 20 - 5.2 = 14.8V$

When $V_{in} \rightarrow -ve$, $|V_{R_{max}}| = |V_{in_{min}} - V_{out}| = |-20 + 0.2| = 19.8V$

$\Rightarrow$ max current flowing through $1k\Omega$ resistor $= \frac{19.8}{1k} = 19.8mA$ When $V_{in} = -20V$

As D_2 is ON when $V_{in} \rightarrow -20V$, If the current is not limited by R , $I_D = I_0 [e^{20/0.025} - 1]$ which will be very very high, burning the diode.

Q14.

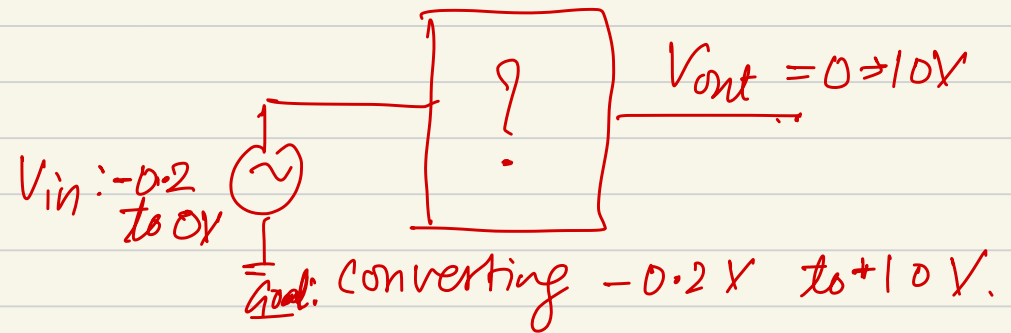


i) for $V_{in} = -0.2V$
 $V_x \rightarrow -ve$, $V_{out} \rightarrow +ve$

$\Rightarrow$ D is OFF (open circuit)

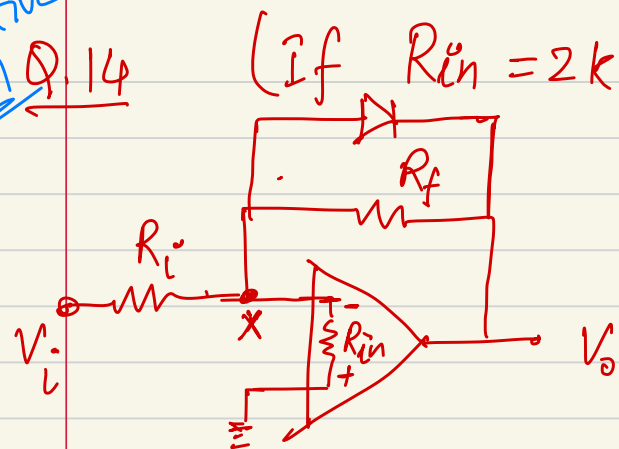
$$\Rightarrow V_{out} = -\frac{R_f}{R_{in}} V_{in} \Rightarrow 10 = +\frac{R_f}{2K} \times 0.2$$

$$R_f = \frac{20K}{0.2} = 100K$$



ii) if $V_{in} = 2V \rightarrow V_x \rightarrow +ve$, $V_{out} \rightarrow -ve$
 $\Rightarrow D_1$ is ON $\Rightarrow V_{out} = V_x \sim 0$

Alternative
Solution Q. 14



(If $R_{in} = 2k\Omega$ is taken as input resistance of op amp)

→ When D is open circuit (OFF)

$$i) \frac{V_i - V_x}{R_i} = \frac{V_x}{R_{in}} + \frac{V_x - V_o}{R_f}$$

$$V_i = -0.2V, V_o = +10V, R_{in} = 2k\Omega$$

$$R_i \& R_f = ?$$

→ $V_x \sim 0V$ as A_o can be assumed to very high.

$$\Rightarrow \frac{V_i}{R_i} = -\frac{V_o}{R_f} \Rightarrow \frac{V_o}{V_i} = -\frac{R_f}{R_i}$$

$$\Rightarrow \frac{+10}{-0.2} = +\frac{R_f}{R_i}$$

$$\Rightarrow \boxed{R_f = 50R_i}$$

ii) Same as previous solution, for a chosen R_i , R_f can be calculated.
 $V_o \sim 0$.